

Abstract Algebra II

Homework 3

Due: 25th March 2026

Submit only Questions 1 and 2. (All fields are assumed to have **characteristic 0**.)

1. Let K be a field and $f \in K[x]$ an irreducible polynomial with Galois group G . (This means that for some splitting field L containing *all* the roots of f , we have $G = \text{Gal}(L/K)$. Note also that splitting field here refers to the *smallest* field containing all the roots of f .) Let $X = \{a_1, \dots, a_n\}$ be the set of distinct roots of f contained in some splitting field L .
 - (a) Show that there exists an injective group homomorphism $G \rightarrow \text{Perm}(X)$, i.e., the group of all permutations of the a_i 's.
 - (b) Show that $n \mid |G|$.
2. Assume the following facts about the polynomial $f(x) = x^5 - 5x - 1 \in \mathbb{Q}[x]$:
 - (i) it has less than 5 real roots;
 - (ii) it is irreducible over $\mathbb{Q}$ (you will see later on how to justify this).

Prove that the Galois group of f is isomorphic to S_5 .

3. Let L be a finite extension of a field K . Show that there exists some $u \in L$ such that $L = K(u)$. (Such a u is called a *primitive element*.)
4. Let $f \in K[x]$ be an irreducible polynomial and L a splitting field of f over K .
 - (a) For an algebraic closure $\overline{K}$ of K containing L and any root α of f , show that every K -embedding $\sigma : K(\alpha) \rightarrow \overline{K}$ extends to an automorphism $\tilde{\sigma} \in \text{Aut}(L/K)$.
 - (b) For any pair of roots α, β of f , show that there exists an automorphism $\tau \in \text{Aut}(L/K)$ such that $\tau(\alpha) = \beta$.
 - (c) Show that L/K is normal, and hence Galois.
5. Let $K = \mathbb{Q}(\sqrt[4]{2}, i)$, we saw that $K/\mathbb{Q}$ is Galois. Compute and describe completely the Galois group $G = \text{Gal}(K/\mathbb{Q})$. (What this means is describe the action of each $\sigma \in G$ on the generators $\sqrt[4]{2}$ and i . This example is classical.)